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Aerodynamics Lab Project

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**Aerodynamic Analysis of
Concorde and Su-57 Nose Sections Using
Schlieren Imaging**

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Abstract

This report presents the study of supersonic flow over scaled nose-section models of the Concorde and Sukhoi Su-57 aircraft at Mach 2.2. Schlieren imaging was used to visualize and quantify shock-waves at angles of attack of 0° , 5° , 8° , and 10° . CAD models were first prepared, and then CNC machining was used to fabricate the nose sections. Shock angles measured from Schlieren images were compared against theoretical predictions using both oblique-shock relations (θ - β - M , applicable to 2D wedge flows) and Taylor-Maccoll conical-flow theory (for 3D axisymmetric bodies).

For the nearly axisymmetric Concorde nose, experimental shock angles had some agreement with Taylor-Maccoll predictions (within ± 2 - 3°). In contrast, the Su-57's nose geometry produced significantly different shocks due to its non-axisymmetric design. The shock has some curvature.

Acknowledgement

We express our sincere gratitude to Prof. Sandeep Saha for his valuable guidance, constant support, and insightful feedback throughout the progress of this work. We are grateful to the Central Workshop (SWISS Lab) for providing access to the CNC and EDM machining facilities, which enabled the precise fabrication of the nose models. We also acknowledge the Mechanical Workshop of the Department of Aerospace Engineering for their assistance with lathing, drilling, and turning operations during the manufacturing process. We would like to extend our special thanks to the lab assistants and technical staff of both the Central Workshop and the Aerodynamics Laboratory for their timely help and expertise during machining and experimental setup.

Finally, we extend our appreciation to the Aerodynamics Laboratory for providing access to the high-speed wind tunnel facilities and for their support in conducting the schlieren experiments, which enabled the successful visualization and analysis of shock structures for our study.

1 Introduction

1.1 Background

while designing supersonic aircrafts the major concerns are **shock-induced wave drag** and **aerodynamic heating**. To study how geometry influences supersonic flow behaviour, two contrasting nose profiles were selected:

- **Concorde** — a long-range civil Mach 2 airliner optimized for efficient cruise.
- **Sukhoi Su-57** — a fifth-generation multirole stealth fighter focused on survivability and agility.

Concorde Nose Geometry

The Concorde features a **long, slender, pointed droop nose** designed to minimize wave drag during cruise. The nose and visor could be lowered during takeoff/landing to improve cockpit visibility, and raised at supersonic speeds to maintain laminar flow and prevent drag rise. The fuselage is slender, with an **ogival-delta wing** and smooth curvature favouring low shock losses and thermally manageable flight.



Figure 1: Concorde – drooped slender nose

Sukhoi Su-57 Nose Geometry

The Sukhoi Su-57 has **chined, faceted nose** and Blended wing–fuselage with all-moving tailplanes, and leading-edge root extensions. The Su-57 is optimized for survivability in contested airspace, combining stealth and high agility. It prioritizes radar cross-section reduction and sustained high- α maneuverability, accepting higher wave drag through chined, faceted nose geometry that deliberately creates specific shock signatures.



Figure 2: Sukhoi Su-57 – chined faceted nose designed for stealth and agility

Why These Two Models Were Chosen

Concorde and Su-57 represent two opposite ends of supersonic aerodynamic design philosophy:

- **Concorde** — an almost axisymmetric, wave-drag-minimized, thermally stable supersonic cruiser.
- **Su-57** — a 3-D chined, stealth-driven geometry enabling high maneuverability at the cost of stronger shock structures.

Studying these nose sections allows direct comparison of shock waves between a drag-optimized civil supersonic aircraft and a maneuverability/stealth-oriented fighter. Our schlieren experiments clearly reveal how geometry alone can determine shock angle, wave strength, and boundary-layer development.

1.2 Objectives

This laboratory project has two primary objectives:

1. Experiment: Use Schlieren imaging to capture shock wave geometry over Concorde and Su-57 nose models across a range of angles of attack (0° – 10°) at Mach 2.2.
2. Theoretical validation: Compare experimental shock angles with predictions from:
 - Oblique shock relations ($\theta - \beta - M$ charts) for 2D wedge baseline
 - Taylor-Maccoll conical flow theory for 3D axisymmetric Concorde nose

2 Model fabrication

2.1 CAD Design

We started by looking for existing CAD designs for the Concorde and Sukhoi Su-57 aircraft. We got a graphics file for these models. From these graphics files, the nose section was separated. The nose sections were then scaled down to appropriate dimensions to fit in the test section of the wind tunnel. These graphics files were then converted into CAD models by taking the cross sections of the model and using the method of lofting to create a clean solid model of the nose.

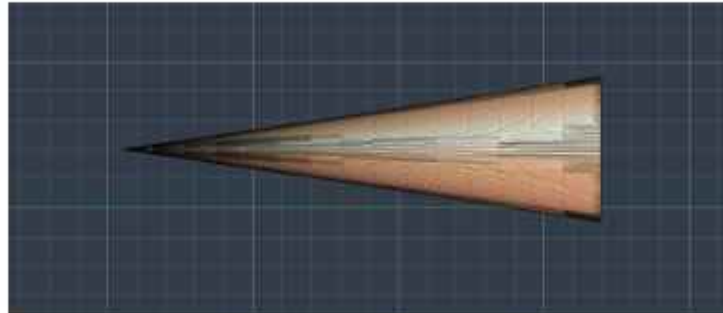


Figure 3: Concorde Nose section Graphics file

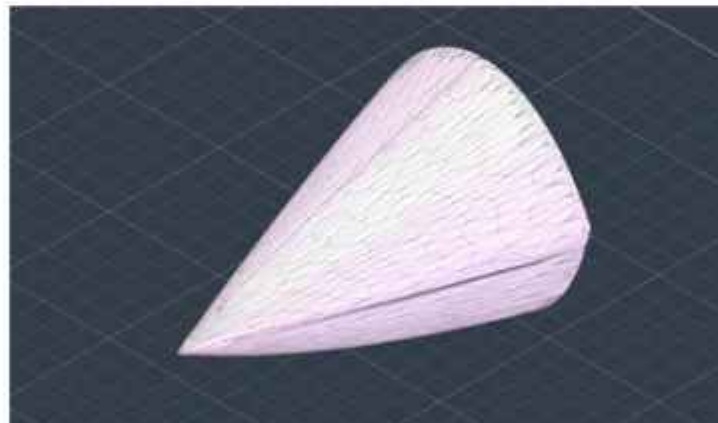


Figure 4: Su-57 Nose section Graphics file

Concorde CAD Model

- Height of the model- 85 mm
- Maximum diameter- 25 mm

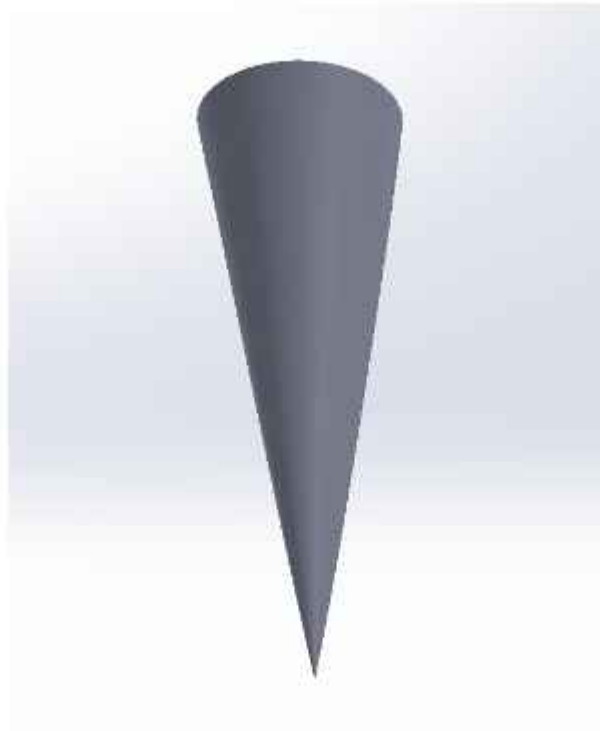


Figure 5: Concorde Nose section CAD

Su-57 CAD model

- Height of the model- 42.5 mm
- Maximum diameter- 30 mm

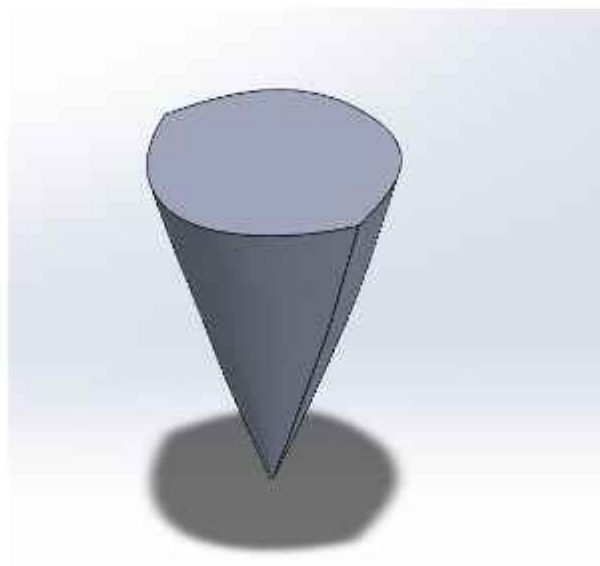


Figure 6: Su-57 Nose section CAD

2.2 Material Preparation

For CNC machining, a strut of the required dimensions was prepared. An additional length of 30 mm was included to allow proper gripping in the machine setup, and an extra 10 mm was provided as a machining buffer.

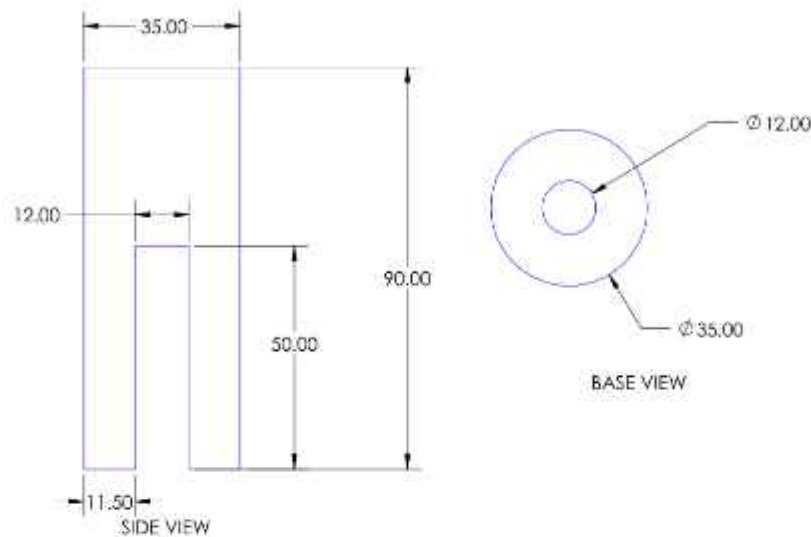


Figure 7: 2D CAD drawings of the strut

Mild steel was selected as the material for the strut due to its ability to withstand the high pressure loads encountered in the high-speed wind tunnel without undergoing deformation. A mild steel beam matching the required maximum diameter was chosen and cut to the specified height.



Figure 8: Cutting of the mild steel beam to the required height.

Facing operations were performed to obtain a smooth and uniform top surface. For mounting the model inside the wind tunnel, a support rod was required. The rod had a diameter of 12 mm and a total length of 8 inches. The end of the rod was turned down to an M10 section over a length of 10 mm.

In the prepared mild steel strut, a hole of diameter 10 mm was drilled to attach the

supporting rod. The drilling depth was set to 50 mm, to account for the 30 mm required for holding the strut in the CNC machine and the additional 10 mm machining buffer.



(a) Lathe machining setup



(b) Drilling of 10 mm diameter hole

Figure 9: Lathe machining and drilling of the 10 mm diameter hole in the strut.

2.3 CNC machining

After Lathing and drilling are done, the specimen is loaded into the 3D CNC (Computer numerical control) with the help of the **Central Workshop (SWISS Lab)**, where the nose section of the CAD models comes into a shape.



(a) Concorde Nose



(b) Su 57 Nose

Figure 10: 3D CNC Setup after smoothing

First, a rough cutting is done in the CNC, and then follows the smoothing surface process.

2.4 EDM

After the CNC machining is completed, the remaining 30-40 mm of support material that holds the specimen during the process is removed using EDM (Electrical Discharge Machining).



(a) Electrical Discharge Machine



(b) EDM Specimen setup

Figure 11: Cutting the base part from the specimen

EDM is chosen because it allows extremely precise cutting without inducing mechanical stresses or deformation on the finished nose section. This ensures the final geometry remains accurate and the surface is not damaged by conventional sawing or machining methods.

2.5 Welding

For the attachment of the supporting rod to the finished specimen, we had two options: **brass welding** and threading. Since the CNC holding position buffers around 10 mm, the drill was not sufficient to make the threading process. Brass welding ensures that the fabricated components remain durable during handling and testing. Its relatively low melting point allows easier weldability with minimal heat distortion, which helps in achieving precise joints without affecting the final geometry of the specimen.



Figure 12: Brass welding the Supporting rod to the Base of the Nose section

Additionally, brass is non-sparking and stable under thermal loads, which prevents any damage to the surface of the specimen.



Figure 13: Final look of Concorde and Su 57 testing specimen

3 Experiment

3.1 Wind tunnel Facility

The wind tunnel used in this experiment is a **blowdown-type supersonic wind tunnel** designed to operate at a Mach number of $M = 2.2$. The settling chamber supplies air at an external pressure of **6 atm**. The test section has a cross-sectional area of **100 mm $\times$ 50 mm**.



Figure 14: Supersonic wind tunnel test section

3.2 Schlieren Setup

Schlieren imaging is an optical flow visualization technique developed in 1864 by **August Toepler** to observe fluid flows by detecting spatial variations in refractive index.



Figure 15: August Toepler — inventor of Schlieren imaging (1864).

When light passes through a medium with density gradients (as in a shock wave), the local refractive index changes, causing the light rays to bend (refract).

Schlieren imaging converts these deflections into visible brightness differences, which help us visualize the density gradient of the flow. Shock waves produce very sharp

density gradients, which deflect light rays. These deflections are converted into intensity variations proportional to the density gradient normal to the knife edge:

$$\Delta I \propto \frac{\partial \rho}{\partial n}$$

where ρ is the density and n is the direction normal to the knife edge.

Light Source & Collimation

A point light source is made parallel via a large parabolic/spherical mirror ($f1-2m$) to produce a parallel light beam through the test section.



Figure 16: Light source and collimation setup.

Test Section

Supersonic flow (Mach 2.2) passes the model, and the formation of shock waves causes density gradients.

Focusing Optics & Knife Edge

A second mirror refocuses light onto a focal plane. A razor blade (knife edge) is positioned precisely at the focal point, blocking approximately half the light beam.



Figure 17: Knife edge placed at focal plane.

Camera

We used our mobile phones to capture the images. Light rays deflected toward the knife edge (positive density gradient) are blocked and appear darker, and the rays deflected away appear brighter.

3.3 Procedure

We started the experiment by placing our model in the test section first at 0 degree angle of attack.



Figure 18: Model placed inside the test section

The position of the lenses was then adjusted so that we could capture the shock waves properly. To align the Schlieren properly, candle flames were used. Candle flames create visible density gradients. The schlieren system visualizes this deflection as a bright/dark pattern around the flame. This can be used to check if the knife edge is positioned correctly. If the flame's plume appears as clear, high-contrast density gradients, then we have positioned correctly. If the setup is misaligned, the image will be blurry or show no contrast. This can be used to align the setup properly before the actual wind tunnel test begins.



Figure 19: Candle used for Schlieren alignment.

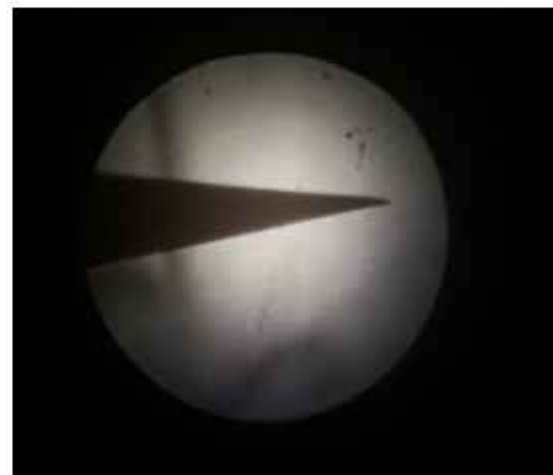


Figure 20: Candle plume visualized as density gradients.

Once setup is done, the compressor is turned on and we wait for the pressure to build up. Meanwhile we setup our camera (we used our mobile phones). Then the lab assistant operates the tunnel and the flow is passed over a short time. We recorded videos of the shock wave.

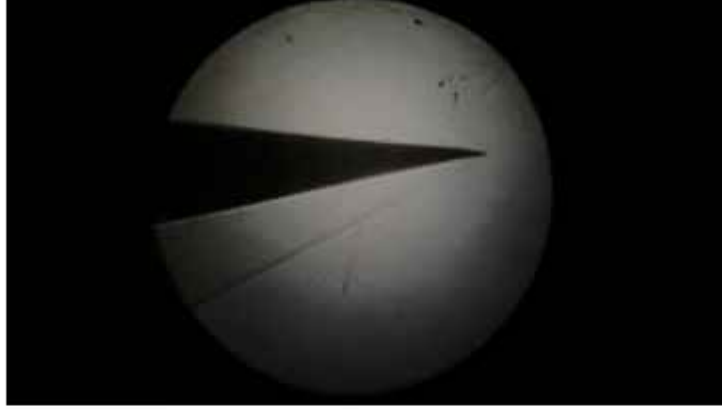


Figure 21: Shock waves captured at 0° angle of attack.

Then the angle is changed to 5, 8 and 10 accurately with the help of protractor placed in the tunnel and the shock waves are recorded.



Figure 22: Protractor used for adjusting angle of attack.

Similar procedure is repeated for Sukhoi Su-57.

3.4 Image processing and Shock angle measurement

From the recorded videos, frames were extracted using a MATLAB code, which is attached in the Appendix section. In these frames, we can clearly see the shock formation and the development of the boundary layer. From these frames, the image showing the complete shock structure was selected. Image contrast and sharpness were varied to make the shock more clear and visible.

Similarly, we have extracted images for each angle of attack for both the models. Then using CAD we have measured and marked the shock angles on the images.

4 Results

4.1 Experimental images

The images display shock structures and boundary layer formation for angles of attack 0° , 5° , 8° and 10° .

4.1.1 Concorde Model

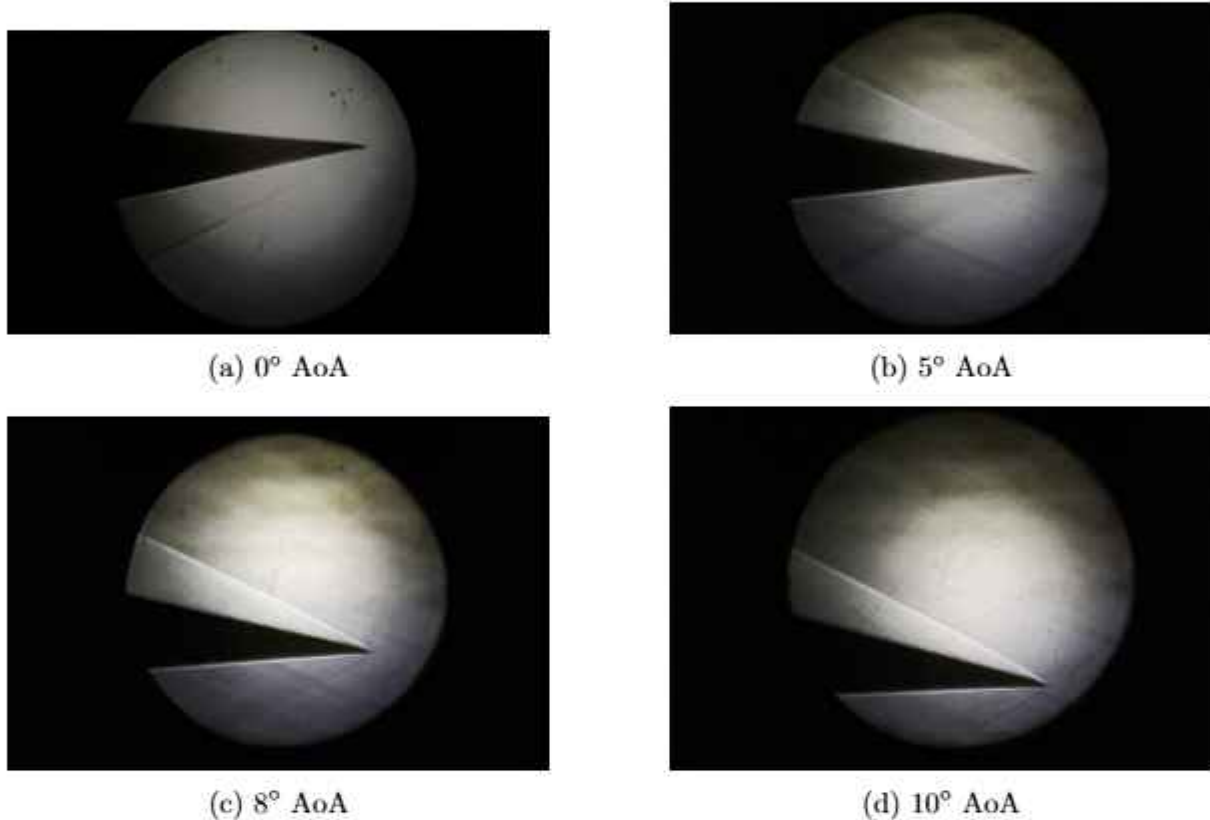


Figure 23: Schlieren visualization of shock waves for Concorde model at different angles of attack.

For the Concorde nose section, we can see that at all angles of attack, we have a straight and attached conical shock (in the image, we can only see two straight lines). We can also observe that the shock on the bottom side of the aircraft nose becomes slightly stronger and closer to the surface (note: the image is flipped and the bottom is the windward side as we give a positive angle of attack). The shock on the bottom is more visible at $\alpha = 10^\circ$.

On the other hand, we can observe that the shock on the top surface becomes weaker and diffuses (we can see it becomes fainter). This mild asymmetry is the first sign of 3D effects, but still much weaker than on the Su-57.

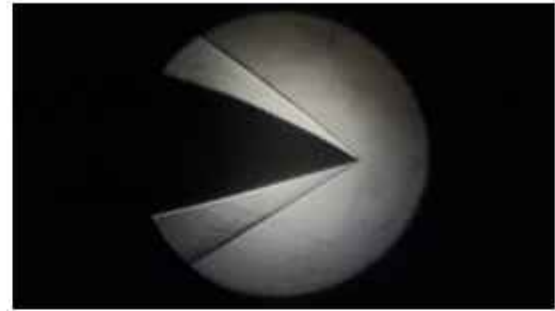
We can also clearly see a thinner boundary layer on the top surface of the nose.

We can also observe some shock reflections from the wall of the test section.

4.1.2 SU-57 Model



(a) 0° AoA



(b) 5° AoA



(c) 8° AoA



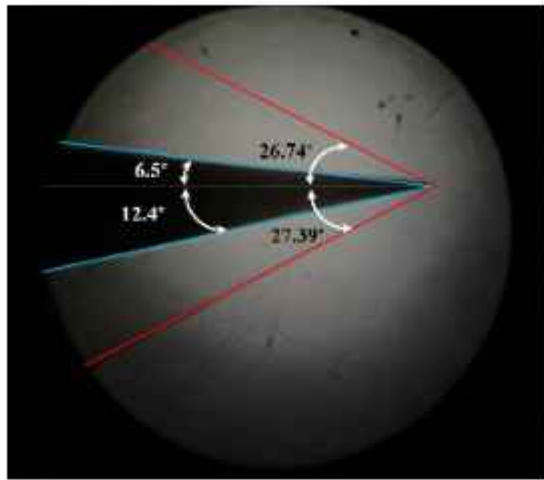
(d) 10° AoA

Figure 24: Schlieren visualization of shock waves for SU-57 model at different angles of attack.

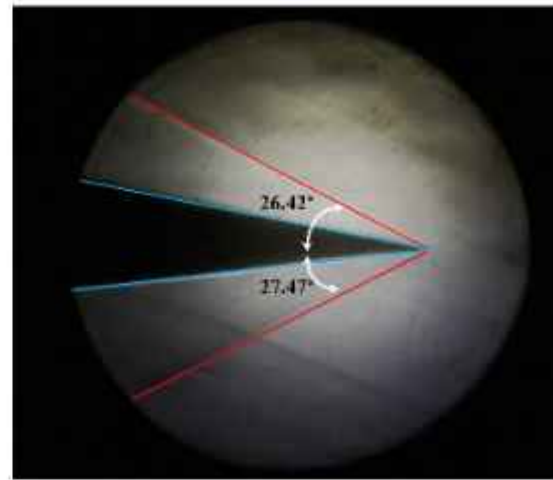
For the Su-57 nose section, we can see that at all angles of attack, we have a curved and attached shock. This indicates strong 3D effects (the shock shape changes as you move around the fuselage circumference). We can also observe that the shock on the bottom side of the aircraft nose becomes slightly stronger and closer to the surface (note: the image is flipped and the bottom is the windward side as we give a positive angle of attack). The shock on the bottom is more visible at $\alpha = 10^\circ$.

On the other hand, we can observe that the shock on the top surface becomes weaker.

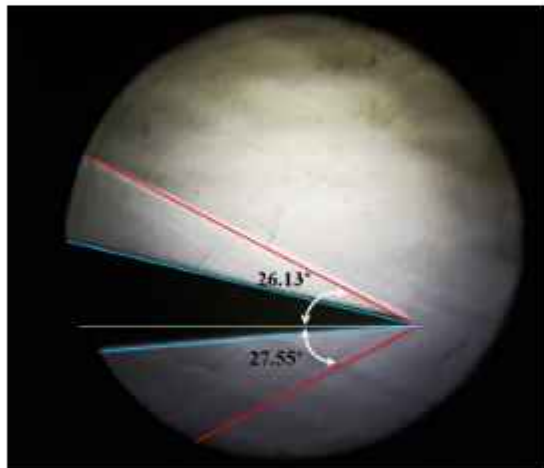
4.2 Shock Angle measurement



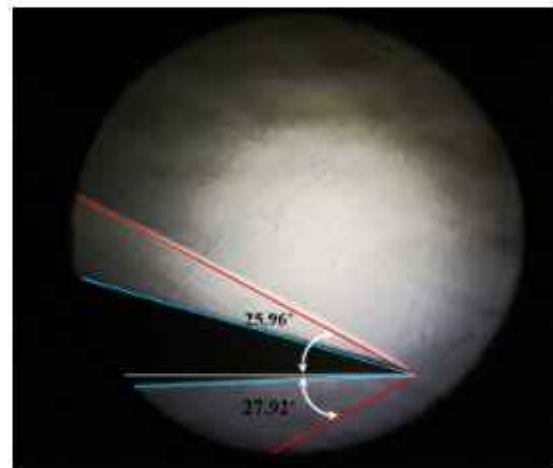
(a) 0° AoA



(b) 5° AoA

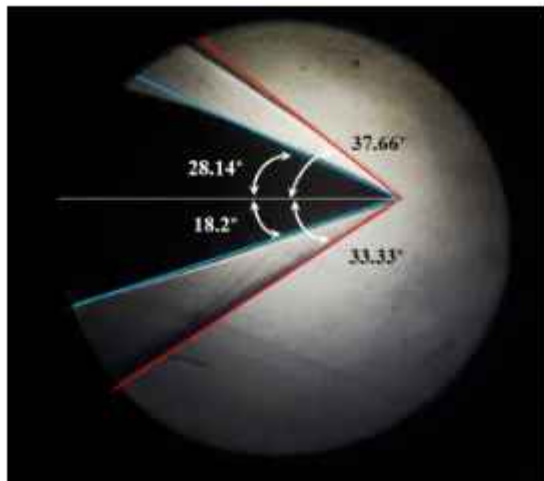


(c) 8° AoA

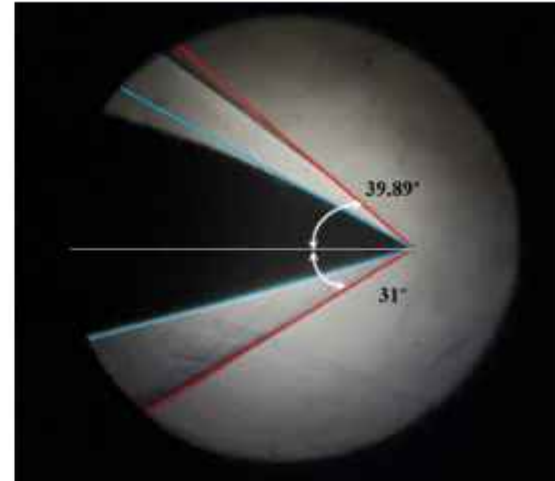


(d) 10° AoA

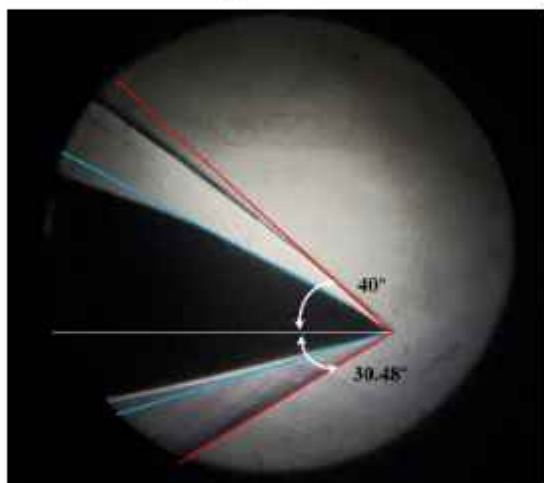
Figure 25: Measured shock angles for Concorde model at 0°, 5°, 8°, and 10° angle of attack.



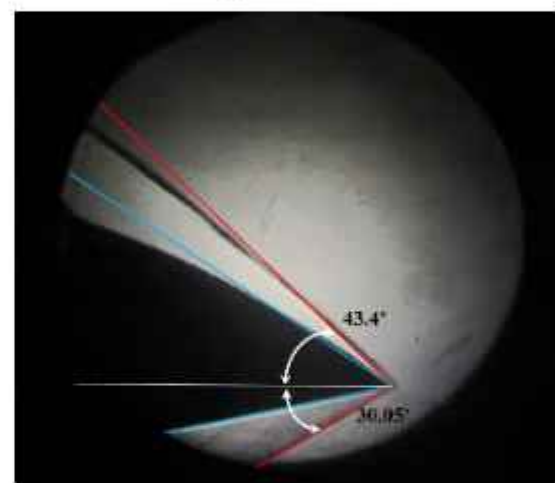
(a) 0° AoA



(b) 5° AoA



(c) 8° AoA



(d) 10° AoA

Figure 26: Measured shock angles for SU-57 model at 0°, 5°, 8°, and 10° angle of attack.

5 Theoretical Analysis

To understand the shock formations at supersonic flow, we analyse the fundamental theories used to predict the flow characteristics. We begin with the simplified two-dimensional θ - β - M relation of oblique shock used for flow over 2D wedges and then more complex Taylor-Maccoll relation for axisymmetric bodies. We compare each of these theories with the experimental results.

5.1 2D Wedge Theory

We consider a 2D wedge with wedge angle measured from CAD of the Concorde nose model and use the θ - β - M relation to find the β (corresponding shock angle)

5.1.1 The θ - β - M equation

Using the continuity equation and the fact that the tangential velocity component does not change across the shock, trigonometric relations eventually lead to the θ - β - M equation, which shows θ as a function of M_1 , β , and γ , where γ is the heat capacity ratio.

$$\tan \theta = 2 \cot \beta \left(\frac{M_1^2 \sin^2 \beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2} \right) \quad (1)$$

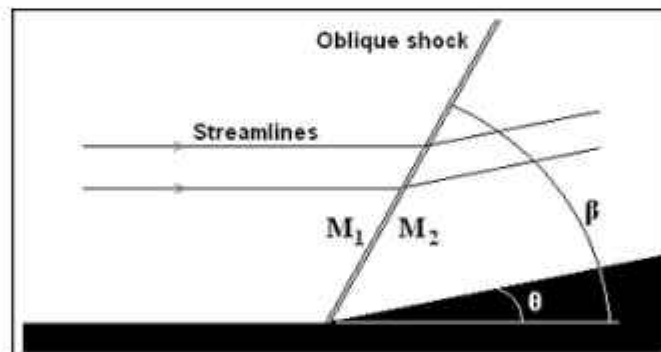


Figure 27: Oblique Shock on a Wedge

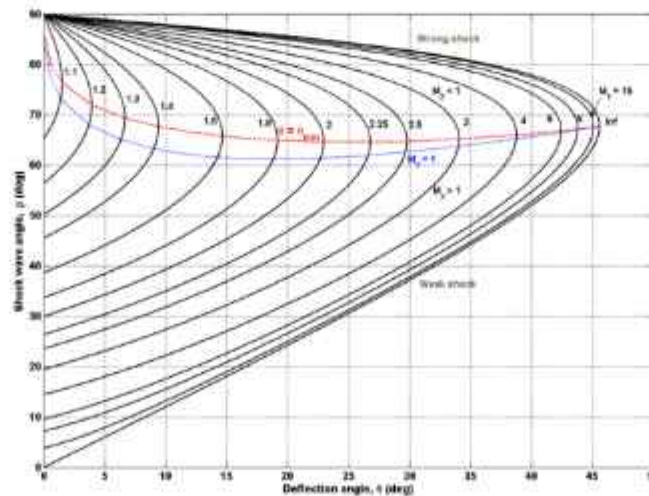


Figure 28: θ - β - M Relation

The Concorde nose can be approximated as a two-stage wedge with flow deflection angles of 6.5° and 12.4° .

From the θ - β - M relation for Mach 2.2 we get $\beta'_1 = 32^\circ$ and $\beta'_2 = 37.87^\circ$. From the experiment image, the corresponding oblique shock angles were found to be $\beta_1 = 26.74^\circ$ and $\beta_2 = 27.39^\circ$. (where, β_1 is for lower surface and β_2 for upper surface)

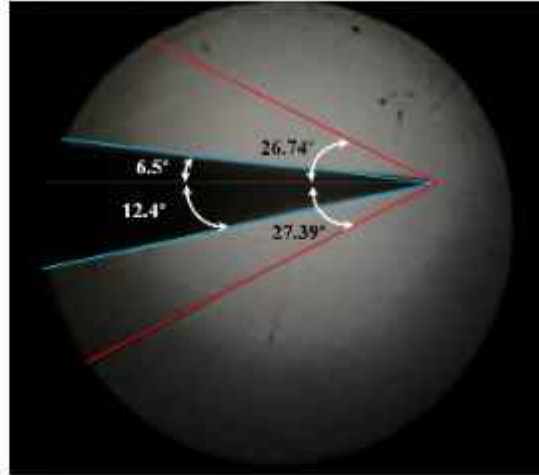


Figure 29: Concorde zero AoA

We can see that for a 2D wedge, for the same Mach number and same deflection, we have a strong oblique shock, with large β . The variation from the experimental data is 5° - 10° . This is due to the 3D relieving effect, where a 3-D body generates weaker shocks (smaller pressure jumps, smaller drag) than a 2-D body with the same local deflection angle, because the flow can “escape” sideways around the body.

5.2 Taylor Macolle

The exact nonlinear solution to the axisymmetric supersonic flow over a sharp cone at zero angle of attack to the free stream is obtained by numerically solving the Taylor–Maccoll equation. Taylor–Maccoll flow is the canonical solution for steady, inviscid, axisymmetric supersonic flow past a sharp cone at zero angle of attack (figure 33).

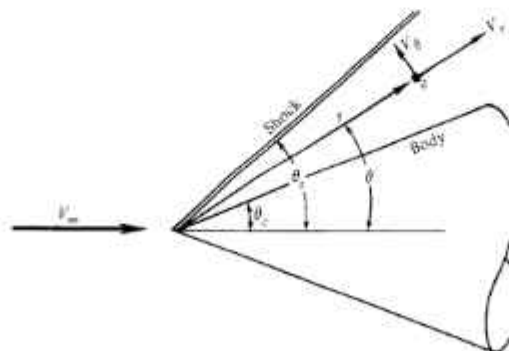


Figure 30: Sharp cone at zero angle of attack

The main assumptions are:

- Flow is steady, inviscid, and adiabatic (perfect gas, typically $\gamma = 1.4$).

-
- The geometry is a right circular cone with semi-vertex (half) angle θ_c .
 - Axisymmetric, conical flow: all flow properties are constant along rays from the apex (depend only on the polar angle, not on radius or azimuth).
 - A single attached conical shock exists whose apex coincides with the cone apex.

With these assumptions, the full three-dimensional Euler equations reduce (in spherical coordinates) to a single ordinary differential equation (the Taylor–Maccoll equation) for the radial velocity component as a function of polar angle.

$$\frac{\gamma - 1}{2} \left[V_{r_{\max}}^2 - V_r^2 - \left(\frac{dV_r}{d\theta} \right)^2 \right] \left[2V_r + \frac{dV_r}{d\theta} \cot \theta + \frac{d^2 V_r}{d\theta^2} \right] - \frac{dV_r}{d\theta} \left[V_r \frac{dV_r}{d\theta} + \frac{dV_r}{d\theta} \left(\frac{d^2 V_r}{d\theta^2} \right) \right] = 0 \quad (2)$$

This ODE is integrated numerically from the shock to the cone surface, with boundary conditions from oblique-shock theory at the shock and a tangency condition (no normal flow) at the cone wall.

The real Concorde nose is an ogive (non-linear) profile rather than a perfect straight-line cone, but it is long, slender, and nearly rotationally symmetric about the aircraft axis. Over the small span of our model nose, cross-sections are almost circular and symmetric and there are no sharp edges as on the Su-57. Thus the forebody flow is close to axisymmetric. The Concorde nose can therefore be approximated as a right circular cone of **semi-vertex angle 12°** , obtained from the CAD nose profile. Under the assumptions of inviscid, axisymmetric, conical flow at zero incidence, the external flow is governed by the Taylor–Maccoll equation, from which the attached shock angle is determined numerically using a MATLAB solver (see reference) to be $\beta_{\text{TM}} = 27.235691^\circ$, which is close to the experimentally observed value.

6 Discussion

6.1 fabrication

The fabrication of the model was a tedious process. Care was taken to ensure that the tip remains sharp. The sharp tip is verified during experiments as we have an attached shock. If the tip were blunt, then we would have a bow shock with a standoff distance.

6.2 Boundary Layer

On the leeward (top) side, the local shock is weaker, so the imposed adverse pressure gradient is smaller and the boundary layer remains relatively thin. Because the static pressure on the leeward side is lower than on the windward side, the boundary layer there has lower density than the outer flow. Across this thin layer, the density changes from a low value at the wall to a higher value in the freestream over a short distance, producing a steep density gradient at the boundary-layer edge. In schlieren, which is sensitive to density gradients, this sharp gradient appears as a clear bright band. On the windward (bottom) side, the boundary layer is thicker, and the density change is spread over a larger distance, giving a gentler gradient and hence lower image contrast. With the knife-edge oriented to detect vertical density gradients, the schlieren system therefore captures the sharp boundary-layer edge on the leeward (top) surface more clearly than on the windward side.

6.3 Three-Dimensional Relieving Effect

In compressible aerodynamics, the three-dimensional (3D) relieving effect means the reduction in shock strength and wave drag that occurs when a two-dimensional flow deflection is achieved on a three-dimensional body. For the same local flow deflection angle, a 3D body (e.g. a cone) produces a weaker shock than a 2D body (e.g. a wedge), because the flow has an additional lateral degree of freedom and can partially “escape” around the body.

For a two-dimensional wedge at freestream Mach number M_∞ , the flow is constrained to turn in a single plane, and a given deflection angle θ produces a relatively strong oblique shock with shock angle β_{2D} and a large pressure ratio p_2/p_1 , as obtained from the θ - β - M relation. For a right circular cone with the same half-angle θ in an otherwise identical flow, the axisymmetric conical flow field allows the fluid to turn not only in the meridional plane but also around the cone. The resulting conical shock has a smaller shock angle $\beta_{3D} < \beta_{2D}$ and a lower post-shock pressure rise, which is the essence of the 3D relieving effect.

In the present study, the Concorde nose is long, slender, and nearly axisymmetric, so its forebody flow can be approximated by that over a right circular cone. Compared to an equivalent 2D wedge of the same effective slope, the conical shock over the Concorde nose is therefore expected to have a smaller shock angle and weaker compression, leading to reduced wave drag at supersonic cruise. This has also been proved experimentally and results are compared in the theoretical analysis.

The Su-57 has sharp chines and faceted surfaces, which break axisymmetry. The flow is no longer that of a simple relieved cone and has multiple interacting shocks and vortices generated by the chines and corners. Although three-dimensionality is still present, it is

used primarily to generate vortex lift and to satisfy stealth shaping requirements, rather than to realise an idealised axisymmetric 3D relieving effect as on the Concorde nose.

6.3.1 Trend Analysis of shock Angle VS AoA

Concorde			
AoA (α)	Lower Shock Angle (β_{lower})	Upper Shock Angle (β_{upper})	Net Change in β
0°	26.74	27.39	N/A
5°	26.42	27.47	0.3
8°	26.13	27.55	0.3
10°	25.96	27.92	0.4
Su 57			
AoA (α)	Lower Shock Angle (β_{lower})	Upper Shock Angle (β_{upper})	Net Change in β
0°	37.66	33.33	N/A
5°	39.89	31	2.3
8°	40	30.48	0.5
10°	43.4	30.05	3.4

Figure 31: Shock angle Data

The shock-angle variation with angle of attack shows two contrasting behaviors for the Concorde and Su-57 nose sections. As the angle of attack increases, the Concorde exhibits only a marginal change in shock orientation, with the upper shock angle decreasing slightly and the lower shock angle increasing by a small amount. This weak dependence on α is due to its smooth ogive-shaped nose, where the curvature allows lateral flow expansion and produces nearly symmetrical compression.

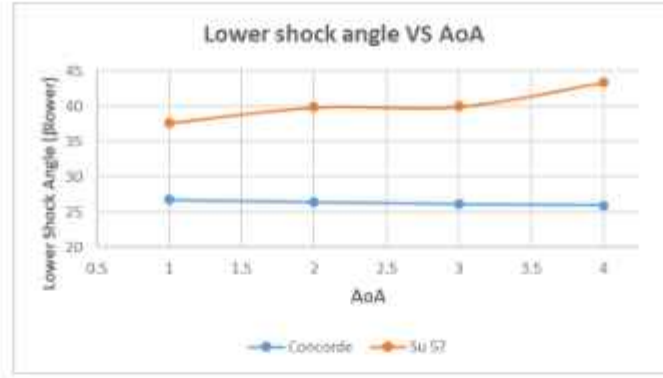


Figure 32: β_{lower} VS AoA

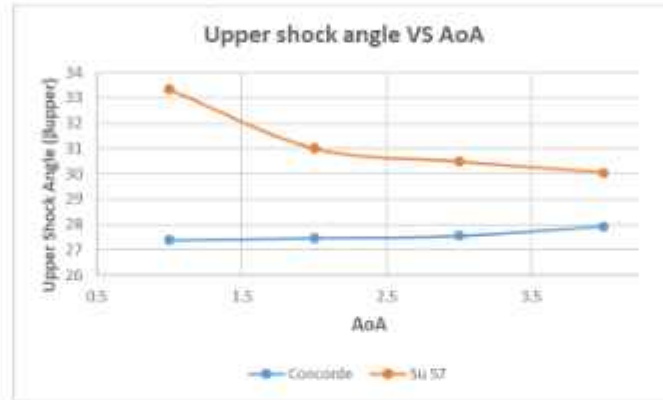


Figure 33: β_{upper} VS AoA

In contrast, the Su-57 nose displays a strong shock response to AoA, where the upper shock angle increases noticeably, while the lower shock angle decreases sharply. This opposite trend arises from its faceted, chine-like geometry, which creates strong asymmetric compression at even small angles of attack. At higher AoA, the sharp chines act as intensified wedge deflections, and the flow interacts with vortical structures that further strengthen the upper shock while reducing compression on the lower side. Consequently, the Su-57 nose exhibits a much higher sensitivity of shock formation to AoA compared to the slender and aerodynamically relieving shape of the Concorde.

7 Conclusions

7.1 Key Findings

1. **Concorde nose validation:** Experimental shock angles measured via Schlieren imaging agree with Taylor–Maccoll conical flow theory within $\pm 2\text{--}3^\circ$, validating the 3D axisymmetric idealization for this geometry.
2. **2D theory inadequacy:** Direct application of 2D oblique shock relations ($\theta\text{--}\beta\text{--}M$) to the Concorde nose overpredicts shock angle by $\sim 5\text{--}10^\circ$, this shows the importance of accounting for 3D relief in conical flows. Shock strength is appreciably weaker in 3D, resulting in lower wave drag and higher post-shock Mach number.
3. **Su-57 non-axisymmetric effects:** The chined geometry of the Su-57 nose produces asymmetric shock and secondary vortex structures absent in the Concorde configuration. This design trades wave-drag efficiency for stealth and high-angle-of-attack agility.
4. **Aerodynamics:** The Concorde–Su-57 comparison shows the fundamental trade-offs in supersonic aircraft design: smooth streamlines optimize efficiency and range; faceted, chined geometries optimize radar cross-section and maneuverability. No single shape is universally optimal.
5. **Boundary layer:** The boundary layer structure is more complex: the windward surfaces show thicker, more disturbed boundary layers due to strong adverse pressure gradients, while leeward surfaces exhibit thinner layers.

7.2 Recommendations for Future Work

1. **High speed cameras:** High-speed cameras and a better schlieren setup can be used to capture more information like the transition of the boundary layer from laminar to turbulent, shock boundary layer interaction, etc.,.
 2. **Extended AoA range:** Measure up to $15\text{--}20^\circ$ to explore shock detachment and boundary-layer separation regimes.
 3. **CFD simulations:** 3D CFD simulations can be done to study the asymmetric shock produced in the Su-57 model.
 4. **Sonic Boom:** We can further study how geometry affects sonic booms and how to optimize the geometry to reduce it.
-

8 References

References

- [1] J. D. Anderson, *Modern Compressible Flow*, McGraw-Hill Education.
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 - [3] "Taylor-Maccoll Solution for Supersonic Flow Over a Cone," MATLAB Central File Exchange, <https://in.mathworks.com/matlabcentral/fileexchange/84490-taylor-maccoll-solution-for-supersonic-flow-over-a-cone>
 - [4] "Supersonic Flow Over a Cone," MATLAB Central File Exchange, <https://in.mathworks.com/matlabcentral/fileexchange/91055-supersonic-flow-over-a-cone>
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Appendix A: MATLAB Code for Frame Extraction

The following script was used to extract individual frames from the schlieren visualization videos. These extracted frames were then processed to observe shock formation, boundary layer development and complete shock structure for different angles of attack.

Listing 1: MATLAB Script used for Frame Extraction from Recorded Videos

```
1 v = VideoReader('Con_10_deg.mp4');
2 v.CurrentTime = 0;           % Start time for frame extraction
3 outDir = 'frames_Con_10_deg';
4
5 if ~exist(outDir, 'dir')
6     mkdir(outDir);
7 end
8
9 k = 0;
10 endTime = 5.0;              % Stop time for frame extraction in
    seconds
11
12 while hasFrame(v) && (v.CurrentTime <= endTime)
13     frame = readFrame(v);
14     fname = fullfile(outDir, sprintf('frame_%05d.png', k));
15     imwrite(frame, fname);
16     k = k + 1;
17 end
```

The extracted frames were later processed to enhance sharpness and contrast for better visualization of shock structures.

Appendix B: Gas table

Table C.2: Weak oblique shock wave angle (β) in degrees for $\gamma = 1.4$

A _h	Flow deflection angle (θ) in degrees																																															
	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46																									
1.92	33.68	31.89	29.79	28.81	27.05	25.49	24.01	22.57	21.17	19.81	18.49	17.21	16.00	14.85	13.75	12.70	11.70	10.75	9.85	9.00	8.20	7.45	6.75																									
1.94	32.71	30.99	28.96	28.01	26.29	24.75	23.30	21.97	20.59	19.26	17.97	16.73	15.54	14.41	13.33	12.30	11.32	10.39	9.51	8.68	7.90	7.16	6.46																									
1.96	32.13	30.42	28.40	27.47	25.78	24.27	22.84	21.53	20.17	18.86	17.59	16.36	15.18	14.05	12.97	11.94	10.96	10.03	9.15	8.32	7.54	6.80	6.10																									
1.98	31.69	29.97	27.97	27.05	25.39	23.90	22.50	21.20	19.85	18.56	17.32	16.11	14.94	13.82	12.75	11.73	10.76	9.84	8.97	8.15	7.38	6.64	5.94																									
2.00	31.65	29.94	27.94	27.02	25.37	23.89	22.50	21.20	19.85	18.56	17.32	16.11	14.94	13.82	12.75	11.73	10.76	9.84	8.97	8.15	7.38	6.64	5.94																									
2.02	31.31	29.60	27.60	26.68	25.04	23.58	22.20	20.92	19.58	18.30	17.07	15.88	14.73	13.62	12.56	11.55	10.58	9.66	8.79	7.97	7.20	6.46	5.76																									
2.04	30.88	29.17	27.17	26.26	24.63	23.19	21.83	20.57	19.24	17.98	16.76	15.58	14.44	13.34	12.29	11.29	10.33	9.41	8.54	7.72	6.96	6.22	5.52																									
2.06	30.66	28.95	26.95	26.04	24.42	22.99	21.64	20.39	19.07	17.82	16.61	15.44	14.31	13.21	12.17	11.17	10.21	9.29	8.42	7.60	6.84	6.10	5.40																									
2.08	30.34	28.63	26.63	25.72	24.11	22.69	21.35	20.10	18.78	17.54	16.34	15.18	14.05	12.95	11.91	10.91	9.95	9.03	8.16	7.34	6.58	5.84	5.14																									
2.10	30.03	28.32	26.32	25.41	23.81	22.40	21.07	19.82	18.51	17.28	16.08	14.93	13.80	12.70	11.67	10.68	9.72	8.80	7.93	7.11	6.35	5.61	4.91																									
2.12	29.73	28.02	26.02	25.11	23.52	22.11	20.79	19.54	18.24	17.01	15.82	14.67	13.54	12.44	11.41	10.42	9.46	8.54	7.67	6.85	6.09	5.35	4.65																									
2.14	29.41	27.70	25.70	24.79	23.21	21.81	20.50	19.25	17.96	16.73	15.55	14.40	13.27	12.17	11.14	10.15	9.19	8.27	7.40	6.58	5.82	5.08	4.38																									
2.16	29.10	27.39	25.39	24.48	22.91	21.51	20.20	18.96	17.67	16.44	15.26	14.11	12.98	11.88	10.86	9.87	8.91	8.00	7.13	6.31	5.55	4.81	4.11																									
2.18	28.87	27.16	25.16	24.25	22.69	21.30	20.00	18.76	17.48	16.25	15.07	13.92	12.79	11.69	10.67	9.68	8.72	7.81	6.94	6.12	5.36	4.62	3.92																									
2.20	28.59	26.88	24.88	23.97	22.42	21.03	19.74	18.50	17.23	16.00	14.82	13.67	12.54	11.44	10.42	9.43	8.47	7.56	6.69	5.87	5.11	4.37	3.67																									
2.22	28.32	26.61	24.61	23.70	22.15	20.77	19.48	18.24	16.97	15.74	14.56	13.41	12.28	11.18	10.16	9.17	8.21	7.30	6.43	5.61	4.85	4.11	3.41																									
2.24	28.06	26.35	24.35	23.44	21.89	20.52	19.24	18.00	16.73	15.50	14.32	13.17	12.04	10.94	9.92	8.93	7.97	7.06	6.19	5.37	4.61	3.87	3.17																									
2.26	27.80	26.09	24.09	23.18	21.63	20.26	18.98	17.74	16.47	15.24	14.06	12.91	11.78	10.68	9.66	8.67	7.71	6.80	5.93	5.11	4.35	3.61	2.91																									
2.28	27.54	25.83	23.83	22.92	21.37	20.00	18.72	17.48	16.21	15.00	13.82	12.67	11.54	10.44	9.42	8.43	7.47	6.56	5.69	4.87	4.11	3.37	2.67																									
2.30	27.29	25.58	23.58	22.67	21.12	19.75	18.47	17.23	15.96	14.74	13.56	12.41	11.28	10.18	9.16	8.17	7.21	6.30	5.43	4.61	3.85	3.11	2.41																									
2.32	27.03	25.32	23.32	22.41	20.86	19.49	18.21	16.97	15.70	14.48	13.30	12.15	11.02	9.92	8.90	7.91	6.95	6.04	5.17	4.35	3.59	2.85	2.15																									
2.34	26.78	25.07	23.07	22.16	20.61	19.24	17.96	16.72	15.45	14.23	13.05	11.90	10.77	9.67	8.65	7.66	6.70	5.79	4.92	4.10	3.34	2.60	1.90																									
2.36	26.53	24.82	22.82	21.91	20.36	18.99	17.71	16.47	15.20	13.98	12.80	11.65	10.52	9.42	8.40	7.41	6.45	5.54	4.67	3.85	3.09	2.35	1.65																									
2.38	26.28	24.57	22.57	21.66	20.11	18.74	17.46	16.22	14.95	13.73	12.55	11.40	10.27	9.17	8.15	7.16	6.20	5.29	4.42	3.60	2.84	2.10	1.40																									
2.40	26.02	24.31	22.31	21.40	19.85	18.48	17.20	15.96	14.69	13.47	12.29	11.14	10.01	8.91	7.89	6.90	5.94	5.03	4.16	3.34	2.58	1.84	1.14																									
2.42	25.77	24.06	22.06	21.15	19.60	18.23	16.95	15.71	14.44	13.22	12.04	10.89	9.76	8.66	7.64	6.67	5.71	4.80	3.93	3.11	2.35	1.61	0.91																									
2.44	25.52	23.81	21.81	20.90	19.35	17.98	16.70	15.46	14.19	12.97	11.79	10.64	9.51	8.41	7.39	6.42	5.46	4.55	3.68	2.86	2.10	1.36	0.66																									
2.46	25.27	23.56	21.56	20.65	19.10	17.73	16.45	15.21	13.94	12.72	11.54	10.39	9.26	8.16	7.14	6.17	5.21	4.30	3.43	2.61	1.85	1.11	0.41																									
2.48	25.02	23.31	21.31	20.40	18.85	17.48	16.20	14.96	13.69	12.47	11.29	10.14	9.01	7.91	6.89	5.92	4.96	4.05	3.18	2.36	1.60	0.86	0.16																									
2.50	24.77	23.06	21.06	20.15	18.60	17.23	15.95	14.71	13.44	12.22	11.04	9.89	8.76	7.66	6.64	5.67	4.71	3.80	2.93	2.11	1.35	0.61	-0.09																									
2.52	24.52	22.81	20.81	19.90	18.35	16.98	15.70	14.46	13.19	11.97	10.79	9.64	8.51	7.41	6.39	5.42	4.46	3.55	2.68	1.86	1.10	0.36	-0.34																									
2.54	24.27	22.56	20.56	19.65	18.10	16.73	15.45	14.21	12.94	11.72	10.54	9.39	8.26	7.16	6.14	5.17	4.21	3.30	2.43	1.61	0.85	0.11	-0.59																									
2.56	24.02	22.31	20.31	19.40	17.85	16.48	15.20	13.96	12.69	11.47	10.29	9.14	8.01	6.91	5.89	4.92	3.96	3.05	2.18	1.36	0.60	-0.14	-0.84																									
2.58	23.77	22.06	20.06	19.15	17.60	16.23	14.95	13.71	12.44	11.22	10.04	8.89	7.76	6.66	5.64	4.67	3.71	2.80	1.93	1.11	0.35	-0.39	-1.09																									
2.60	23.52	21.81	19.81	18.90	17.35	15.98	14.70	13.46	12.19	10.97	9.79	8.64	7.51	6.41	5.39	4.42	3.46	2.55	1.68	0.86	0.10	-0.64	-1.34																									
2.62	23.27	21.56	19.56	18.65	17.10	15.73	14.45	13.21	11.94	10.72	9.54	8.39	7.26	6.16	5.14	4.17	3.21	2.30	1.43	0.61	-0.15	-0.89	-1.59																									
2.64	23.02	21.31	19.31	18.40	16.85	15.48	14.20	12.96	11.69	10.47	9.29	8.14	7.01	5.91	4.89	3.92	2.96	2.05	1.18	0.36	-0.40	-1.14	-1.84																									
2.66	22.77	21.06	19.06	18.15	16.60	15.23	13.95	12.71	11.44	10.22	9.04	7.89	6.76	5.66	4.64	3.67	2.71	1.80	0.93	0.11	-0.65	-1.39	-2.09																									
2.68	22.52	20.81	18.81	17.90	16.35	14.98	13.70	12.46	11.19	9.97	8.79	7.64	6.51	5.41	4.39	3.42	2.46	1.55	0.68	-0.14	-0.88	-1.62	-2.32																									
2.70	22.27	20.56	18.56	17.65	16.10	14.73	13.45	12.21	10.94	9.72	8.54	7.39	6.26	5.16	4.14	3.17	2.21	1.30	0.43	-0.39	-1.13	-1.87	-2.57																									
2.72	22.02	20.31	18.31	17.40	15.85	14.48	13.20	11.96	10.69	9.47	8.29	7.14	6.01	4.91	3.89	2.92	1.96	1.05	0.18	-0.64	-1.38	-2.12	-2.82																									
2.74	21.77	20.06	18.06	17.15	15.60	14.23	12.95	11.71	10.44	9.22	8.04	6.89	5.76	4.66	3.64	2.67	1.71	0.80	-0.07	-0.81	-1.55	-2.29	-2.99																									
2.76	21.52	19.81	17.81	16.90	15.35	13.98	12.70	11.46	10.19	8.97	7.79	6.64	5.51	4.41	3.39	2.42	1.46	0.55	-0.38	-1.12	-1.86	-2.60	-3.30																									
2.78	21.27	19.56	17.56	16.65	15.10	13.73	12.45	11.21	9.94	8.72	7.54	6.39	5.26	4.16	3.14	2.17	1.21	0.30	-0.57	-1.31	-2.05	-2.79	-3.49																									
2.80	21.02	19.31	17.31	16.40	14.85	13.48	12.20	10.96	9.69	8.47	7.29	6.14	5.01	3.91	2.89	1.92	0.96	0.05	-0.82	-1.56	-2.30	-3.04	-3.74																									